

Dynamic Programming Patterns & Self-Balancing Trees

Subject: CS301 | Unit 5: DP & Advanced Trees

1. Executive Overview & Syllabus Scope

Lecture notes on 0/1 Knapsack, Longest Common Subsequence LCS, Matrix Chain Multiplication, AVL Tree rotations LL, RR, LR, RL

2. Key Architectural Invariants & Core Topics

â€¢ Lecture Modules & Key Topics

- Unit 5: Dynamic Programming Patterns & Self-Balancing Trees
- 1. Dynamic Programming Paradigms: Optimal Substructure and Overlapping Subproblems
- 2. 0/1 Knapsack: Recursive recurrence relation, Memoization table, and Tabulation state transition
- 3. Longest Common Subsequence LCS: Table filling and traceback path recovery
- 4. Matrix Chain Multiplication: Parenthesization cost minimization in $O(n^3)$
- 5. AVL Trees: Balance factor definition and 4 rotation cases LL, RR, LR, RL
- 6. Red-Black Trees: Invariant properties Black-height, Red-parent rule and operations

3. University Exam Tips & High-Yield Checklist

- * Practice numerical problems and time complexity derivations under timed conditions.
- * Draw neat labeled block diagrams for full credit in subjective university evaluations.
- * Use AI_Book Quiz & Summary features to test active recall before mid-terms and finals.